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Driving Operational Excellence Through Data Standardization and Integrated Analytics: Delivering Digital Transformation in the North & West Kuwait Asset

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Abstract

Achieving operational excellence in oil & gas operations increasingly hinges on delivering high-quality, consistent, and accessible field data. Historically, fragmented systems and inconsistent reporting structures across the North & West Kuwait Asset impeded optimization efforts. This initiative was launched to unify operational datasets, enforce standardized governance, and implement an integrated analytics platform, enabling real-time, data-driven decision-making and supporting the asset's broader digital transformation, resilience, and sustainability objectives.

The initiative was executed through a structured two-phase methodology. Phase 1 involved a Data Standardization Campaign, during which over 150 critical operational KPIs were meticulously mapped, validated, and standardized through extensive cross-functional stakeholder engagement, establishing a unified data governance framework across multidisciplinary teams. Each KPI, previously reliant on disparate manual processes, was redefined with standardized calculation methodologies and automated primary data sources. In Phase 2, an Integrated Analysis Dashboard was engineered using Spotfire, dynamically integrating real-time feeds from AVOCET and associated platforms. Embedded data transformation scripts ensured integrity, while auditing logic and predictive analytics capabilities enabled proactive field management.

Deployment of the Integrated Analysis Dashboard delivered measurable and transformative improvements in operational performance and data governance across the asset. Data consistency compliance improved by approximately 92%, reflecting the successful consolidation of previously fragmented datasets into a unified, automated environment. Reporting cycle times were shortened by 38%, accelerating the dissemination of operational insights and enabling more agile, informed management responses. Intervention planning timelines improved by 25%, allowing faster execution of field activities and optimized resource utilization.

Real-time visibility into production bottlenecks, steam injection efficiency, and well performance anomalies empowered technical and operational teams to implement corrective actions with greater speed and precision. These enhancements not only optimized daily operations but also strengthened long-term production planning and forecasting capabilities. Moreover, the initiative catalyzed a cultural shift toward

data-driven collaboration, with multidisciplinary teams adopting a more proactive, analytically informed approach to field management. The dashboard's modular, scalable architecture ensures adaptability across additional fields, supporting the asset's ongoing digital evolution and operational excellence initiatives.

Overall, this project demonstrates that integrating standardized operational data into a centralized, intelligent analytics platform can tangibly improve heavy oil asset performance while accelerating broader digital transformation goals within the upstream sector.

This project advances the industry's understanding of scalable data standardization and real-time operational analytics within heavy oil field management. Unlike conventional dashboards, it seamlessly integrates standardized historical data, live KPIs, and forecast analytics into a fully auditable, dynamic environment enhanced with geospatial visualization. The solution presents a replicable, future-proof model for upstream digital transformation, directly enabling enhanced operational decision-making, resource optimization, and contributions to the broader energy transition agenda.

Introduction

High-quality, consistent operational data is foundational to modern oilfield performance—particularly as upstream assets contend with rising complexity, regulatory expectations, and decarbonization targets. Industry analyses show that 60–70% of upstream performance gaps stem from poor data accessibility and standardization (McKinsey & Company, 2023), underscoring the value of structured, governed data. In heavy oil environments, where operations are nonlinear and enhanced oil recovery (EOR) mechanisms such as steam injection add further variability, even small inconsistencies in operational data can delay interventions, hinder optimization, and reduce resource efficiency.

Historically, Heavy Oil – North Kuwait (NKHO) operated within a fragmented data landscape, with siloed workflows, disconnected spreadsheets, offline trackers, and isolated databases. These disjointed systems led to inconsistent KPI definitions, duplicated efforts, and poor interoperability between surface and subsurface domains. The absence of standardized data governance further amplified discrepancies, lowered confidence in reported metrics, and slowed the delivery of actionable insights.

Recognizing that operational excellence depends on trusted, accessible, and consistently defined data, a cross-functional initiative was launched to deliver a data standardization campaign and an integrated analytics environment. This initiative supports Kuwait Vision 2035 and Kuwait Petroleum Corporation's (KPC) digital transformation agenda, aiming to modernize upstream workflows and enable scalable, intelligent operations.

Building on the strategic framework presented in the NKHO Digital Transformation Roadmap (Al-Ballam & Ziyab, 2025), this paper outlines the two-phase implementation:

1. Standardization and automation of over 150 critical operational KPIs through cross-disciplinary stakeholder engagement.
2. Development and deployment of a real-time Integrated Analysis Dashboard in Spotfire, incorporating embedded data transformation logic, audit protocols, and predictive analytics.

By consolidating fragmented data into a governed, intelligent platform, the initiative has strengthened data governance, shortened reporting cycles, and improved intervention planning agility. Beyond the technical outcomes, it has fostered a culture of analytics-driven collaboration—laying the foundation for future digital innovation in NKHO's heavy oil operations.

Background and Challenges

Prior to this initiative, NKHO relied heavily on manual, team-specific processes—spreadsheets, offline logs, and isolated databases—that fragmented operational data and limited collaboration between surface and subsurface teams. The absence of standardized data governance meant that KPI definitions, calculation methods, and update frequencies often varied between groups, leading to inconsistent reporting and duplicated effort.

These inconsistencies extended to critical workflows such as well performance analysis, steam injection planning, and intervention prioritization. Even routine monitoring required time-consuming validation and reconciliation, reducing confidence in reported figures and delaying decision-making. As a result, field management tended to be reactive rather than proactive, with operational adjustments often made after issues had already impacted performance.

This fragmented approach also slowed KOC's broader digital transformation objectives. Without clean, consistently defined, and readily accessible data, the deployment of predictive analytics, cross-functional reporting, and automation was severely constrained—limiting operational agility and delaying progress toward digital maturity, resilience, and sustainability targets.

Collectively, these challenges underscored the urgent need for a unified, governed data environment —one capable of integrating standardized KPIs, enabling real-time analytics, and supporting automated, auditable reporting. Addressing these foundational issues was essential not only to improve day-to-day performance but also to unlock long-term value creation, workforce efficiency, and alignment with KOC's strategic digital transformation roadmap.

Project Objectives and Scope

Project Objectives

The NKHO Data Standardization and Dashboard Development Initiative was launched to translate the strategic recommendations of the 2025 NKHO Digital Transformation Roadmap (Al-Ballam & Ziyab, 2025) into field-level execution. While the roadmap emphasized governance, integration, and forecasting, this initiative operationalized those pillars through a standardized KPI framework and an integrated analytics platform.

The initiative aligned with Kuwait Oil Company's digital transformation strategy and focused on resolving long-standing data fragmentation challenges by:

1. Institutionalizing KPI Standardization – Redefining and validating 150+ operational KPIs across production, planning, and engineering functions. Each KPI was mapped to authoritative data sources and harmonized through a standardized calculation methodology to align with corporate reporting frameworks.
2. Deploying a Real-Time Analytics Platform – Implementing Spotfire as the core visualization and analytics tool, with dynamic integration to AVOCET and embedded analytical capabilities for operational insight.
3. Enabling Proactive Decision-Making – Incorporating automated audit logic, data transformation layers, and predictive analytics to shift field management from reactive response to insight-driven operations.

Scope of Work

The initiative targeted four high-impact operational domains within NKHO:

- Oil Production – Standardized daily, monthly, and cumulative production tracking, including variance analytics and performance diagnostics.
- Steam Injection – Monitoring of planned vs. actual injection volumes, steam-oil ratio (SOR) analytics, and thermal efficiency assessment.
- Well Status and Interventions – Real-time tracking of well closures, intervention planning, and post-activity production gain evaluation.

- Production Losses and Deferment – Categorization and trend analysis of unplanned losses by root cause and impact duration.

The initial deployment covered the South Ratqa (SR) and Umm Niqa (UN) fields, with configurations for both field-specific and combined views. The architecture was designed to be modular and scalable, enabling rapid expansion to additional fields and KPI categories, and serving as a foundational enabler for KOC's long-term digital transformation goals.

Methodology and Execution Approach

This initiative was delivered in two distinct but interconnected phases. Phase 1 focused on establishing a robust foundation of data standardization, while Phase 2 translated this foundation into a fully operational, real-time analytics platform. Each phase was guided by structured stakeholder engagement and executed through agile development cycles.

Phase 1: Data Standardization Campaign

KPI Identification, Mapping, and Validation. Over 150 key performance indicators (KPIs) were identified in collaboration with cross-disciplinary stakeholders from Production Operations, Reservoir Studies, Field Development, Maintenance, and Surveillance. The effort involved auditing legacy trackers, Excel spreadsheets, and isolated reporting tools. Each KPI was prioritized based on operational criticality and value to decision-making, with low-impact or redundant metrics removed.

Strategic data fragmentation challenges highlighted in the NKHO digital roadmap (Al-Ballam & Ziyab, 2025) guided the consolidation effort. KPI validation workshops were conducted to align definitions, reconcile inconsistencies in formula logic and update frequencies, and ensure standardization across measurement units and thresholds. This process directly addressed historical data conflicts that previously hindered operational alignment.

Cross-Functional Engagement and Collaboration. Through iterative working sessions, KPIs were formally endorsed by domain-specific data owners. This engagement fostered collective accountability and strengthened a culture of governance, transparency, and shared responsibility. Previously siloed teams collaborated to define unified performance metrics and ensure their consistent application.

Definition of Data Sources and Lineage. For each KPI, primary and secondary data sources were clearly documented. This included origin systems such as AVOCET, WSMS, and ALMS, as well as data update cadence and ownership responsibilities. Where automation was not possible, placeholders and flags were built into the process to support future system integration and reduce reliance on manual inputs.

Implementation of a Data Governance Framework. To institutionalize the results, a fit-for-purpose data governance framework was rolled out. It included:

- Assignment of data owners and focal points.
- Defined update schedules and quality control mechanisms.
- Business domain classification (e.g., production, injection, losses).
- Introduction of audit protocols and logic checks.

All final definitions were consolidated into a master Data Standardization & Consistency (DSC) Template. This served as the authoritative reference for downstream dashboard development and continues to function as a living document within NKHO's digital operations framework.

Phase 2: Technical Development of Integrated Analysis Dashboard

Following the establishment of standardized KPIs and a unified data governance framework in Phase 1, the second phase focused on the technical development and deployment of a real-time analytics dashboard. This phase was centered on operationalizing the standardized data environment through intelligent visualizations, dynamic reporting, and embedded analytics—transforming raw data into actionable insights.

Platform Selection and Configuration. Spotfire was selected due to its compatibility with KOC's digital standards and support for real-time analytics (TIBCO, 2024). The platform enabled:

- Seamless integration with AVOCET via validated Information Links from the centralized KOC library.
- Live data feeds for critical operational KPIs such as production volumes, losses, well closures, and post-intervention gains.
- Scalable architecture with embedded R/Python environments to support future predictive and diagnostic analytics.

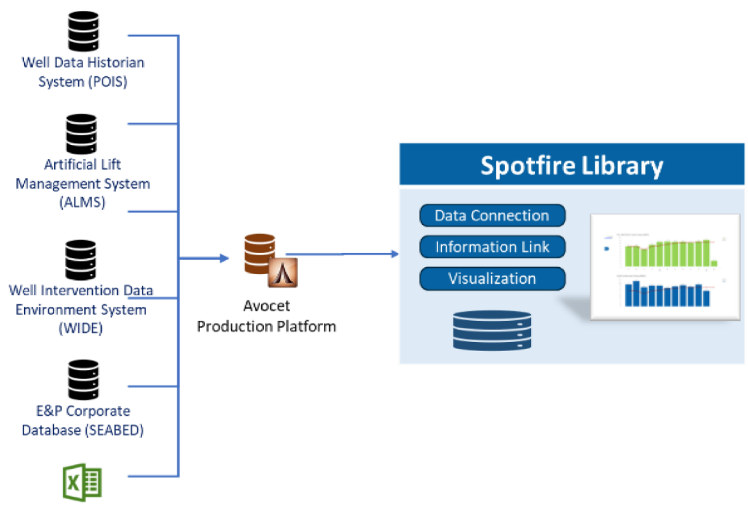

Figure 1—System flow diagram and Spotfire library integration path

Backend Automation and Quality Control. To ensure data integrity, clarity, and real-time usability, the NKHO DSC Dashboard integrates a robust backend automation layer with embedded quality control mechanisms. This technical foundation enables seamless transformation of raw operational data into standardized, actionable insights.

Automated transformation logic was applied to harmonize key metadata across input sources. For example, well block identifiers were standardized (e.g., standardizing names: well, pattern, and block identifiers, etc) to ensure consistency across naming conventions. Calculated columns were introduced to derive key parameters—such as "FIELD" from well names—allowing for structured aggregation and analysis. Additionally, embedded scripts dynamically compute daily, monthly, and annual totals for key metrics like oil, water, and steam volumes, enabling time-sliced analysis without manual intervention.

Functional Overview of Dashboard Modules and User Value. To enhance operational transparency and user engagement, the NKHO DSC Dashboard was structured into modular pages, each tailored to address a specific domain within heavy oil production management. Table 1 summarizes the purpose, key visualizations, and operational value of each dashboard module. This functional overview highlights how the platform enables end users across all stakeholders to access actionable insights, streamline reporting, and support real-time field decision-making across domains such as production, steam injection, losses, interventions, and planning. By linking technical metrics to practical field workflows, the dashboard moves beyond a reporting tool to serve as a core driver of performance optimization.

Table 1—Functional Overview of NKHO DSC Dashboard Modules

Dashboard Page	Purpose	Key Visualizations	End-User Value
Home Page	Provide a high-level overview of key KPIs and updates	KPI summary table, news/ notes section, contact info	Central access point; supports quick situational awareness and user navigation
Oil Production Dashboard	Monitor total oil production trends across various timeframes	Bar charts (daily/monthly/ yearly), production summary tables, field/well filters	Enables performance tracking, historical trend analysis, and production diagnostics
Forecast Information Dashboard	Compare actual vs. forecasted oil and water production	Bar/line graphs for oil and water rate forecasts with filters	Assists in forecast variance analysis and proactive production planning
Water Production Dashboard	Track water production volumes and water cut percentage	Water rate charts, water cut line graphs, filterable production tables	Supports water management strategies and reservoir health monitoring
Steam Injection Dashboard	Analyze steam injection rates and performance	Daily/monthly/yearly steam volume graphs, SOR calculations, annotated bar charts	Assists in thermal efficiency evaluation and steam optimization planning
Oil & Steam Overlay Dashboard	Evaluate the relationship between steam injection and oil production	Comparative graphs of steam/ oil rate and SOR trends	Helps assess steam-to-oil efficiency and identify operational imbalances
Well Closure Dashboard	Monitor closed wells by reason, duration, and field	Bar charts (closure count), pie chart (closure reasons), closure duration graphs	Supports intervention prioritization, root cause analysis, and downtime reduction
Production Loss Dashboard	Diagnose causes and trends in unplanned production losses	Bar/pie charts of loss volumes and causes, historical data tables	Enables rapid root cause analysis and loss mitigation planning
Production Gain Dashboard	Assess incremental gains from interventions	Bar charts of P50 gains per well, detailed intervention result tables	Evaluates intervention effectiveness and supports decision-making for future activities
Activity Tracking Dashboard	Compare planned vs. actual field activities	Bar charts (planned vs. actual), deviation line graph, activity summary tables	Identifies execution gaps and supports performance tracking of operational planning
Geographical Oil Production Dashboard	Visualize oil production spatially across fields	Interactive field maps, production site data tables	Supports spatial performance analysis and resource planning
Geographical Steam Injection Dashboard	Visualize steam injection distribution and patterns	Interactive injection site maps, field/pattern/well-level filters	Enables targeted thermal recovery analysis and steam deployment assessment

A defining capability of the NKHO DSC Dashboard is its Executive KPI Summary page, which integrates more than a dozen critical production, injection, intervention, and downtime metrics into a single, real-time operational view. By replacing numerous static, siloed reports with an interactive, consolidated interface, it provides immediate situational awareness to asset managers and technical teams. This unified view not only streamlines performance monitoring but also accelerates cross-functional decision-making by ensuring that all stakeholders operate from the same, up-to-date operational intelligence (Fig. 2).

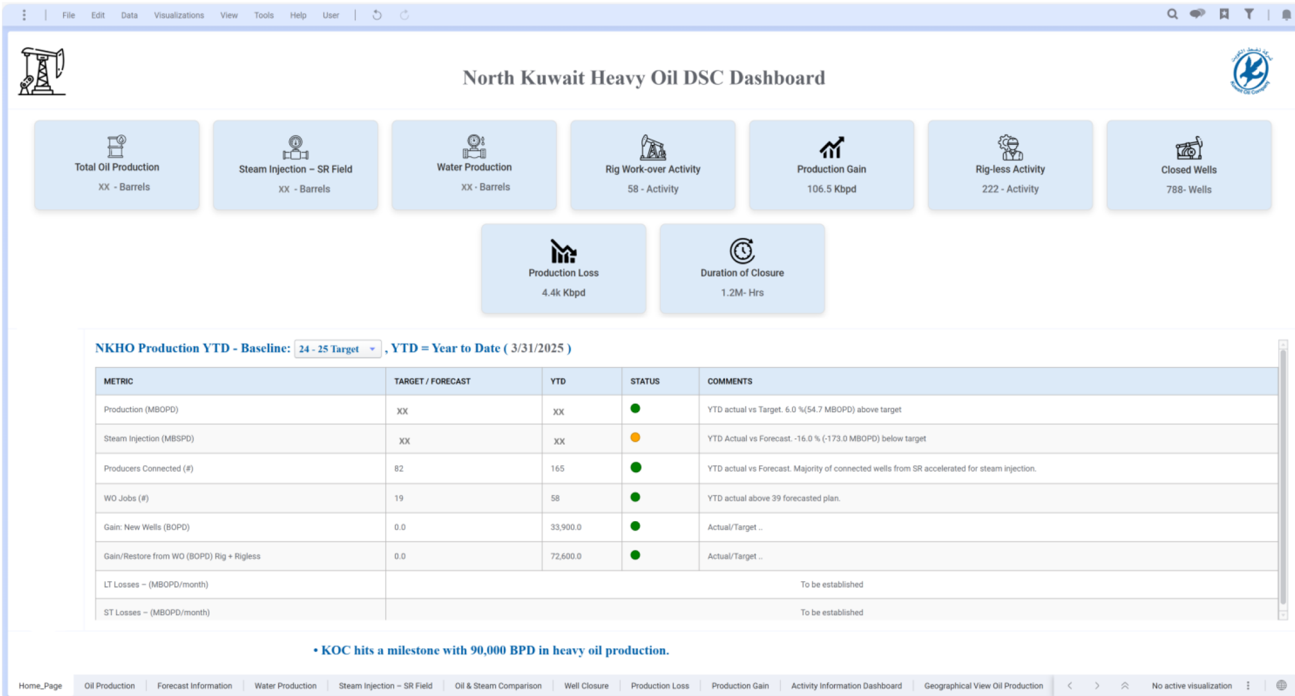

Figure 2—Executive KPI Summary page providing a unified, real#time view of NKHO performance, replacing multiple legacy reports with a single operational intelligence interface

Predictive Analytics and Geospatial Visualization Features

To further support proactive field management and strategic planning, the NKHO Integrated Analysis Dashboard was enhanced with predictive analytics and geospatial visualization capabilities. These features enable users to dynamically assess performance trends, benchmark thermal efficiency, and spatially analyze operational behavior across the asset.

Forecast Comparison and Deviation Monitoring. The dashboard incorporates color-coded bar and line chart visualizations to compare actual versus forecasted oil and water production. These predictive modules allow users to track deviations over daily, monthly, and annual timeframes, highlighting underperformance or overproduction events that may require operational adjustments. This capability enhances visibility into variance trends and supports more agile decision-making in production planning and intervention scheduling. The Forecast Information module compares actual vs. forecasted oil and water rates, using variance thresholds to automatically highlight deviations that require operational review. For example, in mid-2024, the system flagged a rising water variance early, prompting targeted steam allocation adjustments that prevented further performance loss (Fig. 3). This capability transforms forecasting from a static planning exercise into a dynamic, operational control tool.

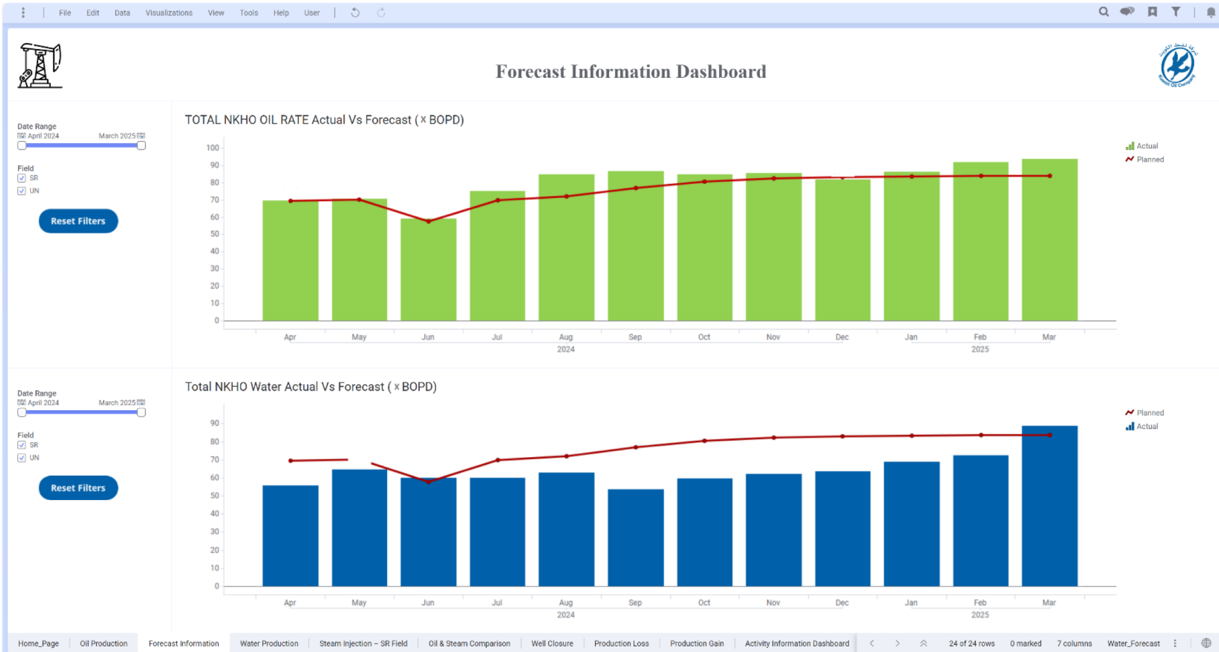

Figure 3—Forecast vs. Actual Performance Tracking

Geospatial Visualization for Operational Insight. Geospatial features are enabled through the integration of XY-coordinate data from the well_location table, joined with asset metadata (e.g., field, block, and pattern classifications). Interactive map layers display spatial distributions of production and injection wells across South Ratqa (SR) and Umm Niqa (UN), providing a visual interface to:

- Identify high- or low-performing zones,
- Track well statuses,
- Diagnose localized inefficiencies.

Fig. 4 and 5 illustrate the dashboard's interactive maps for oil production and steam injection activity, respectively. Users can filter by operational domain, time range, or field region to conduct targeted analysis. NKHO concerned teams routinely apply the steam injection map to validate injection coverage across patterns and adjust deployment based on evolving well-level performance.

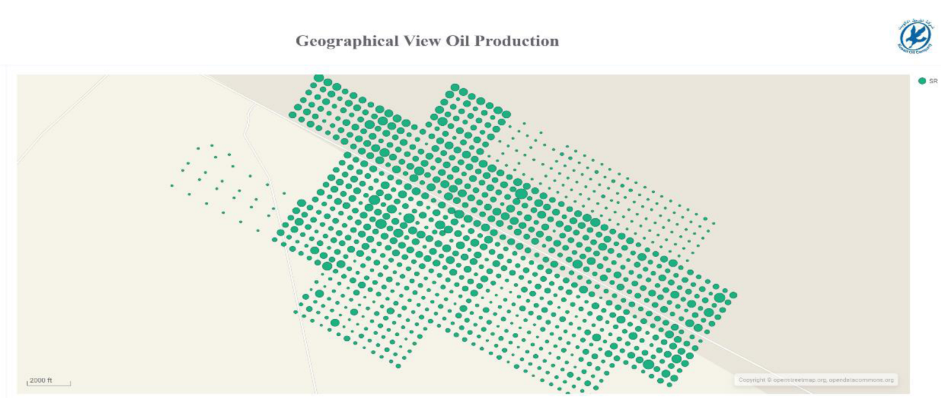

Figure 4—Geographical View Oil Production

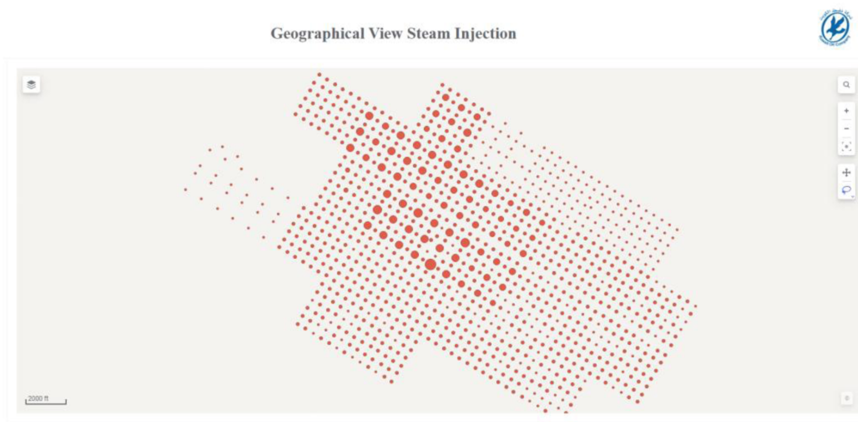

Figure 5—Geographical view steam production

Steam-Oil Ratio (SOR) Overlay and Thermal Efficiency Monitoring. The Oil–Steam Overlay module combines steam injection rates, oil production rates, and calculated Steam-Oil Ratios (SOR) to deliver a precise diagnostic of thermal recovery efficiency. High-SOR zones — indicative of suboptimal steam utilization — are visually flagged within the dashboard, enabling reservoir engineers to quickly target areas requiring operational intervention. For example, in early 2025, the module identified elevated SOR in Pattern B-05, prompting a targeted steam reallocation that improved recovery efficiency by 12% within one month. By directly linking steam performance to production outcomes, the overlay supports rapid, evidence-based adjustments to injection strategies, enhancing thermal efficiency, improving recovery rates, and reducing steam wastage (Fig. 6).

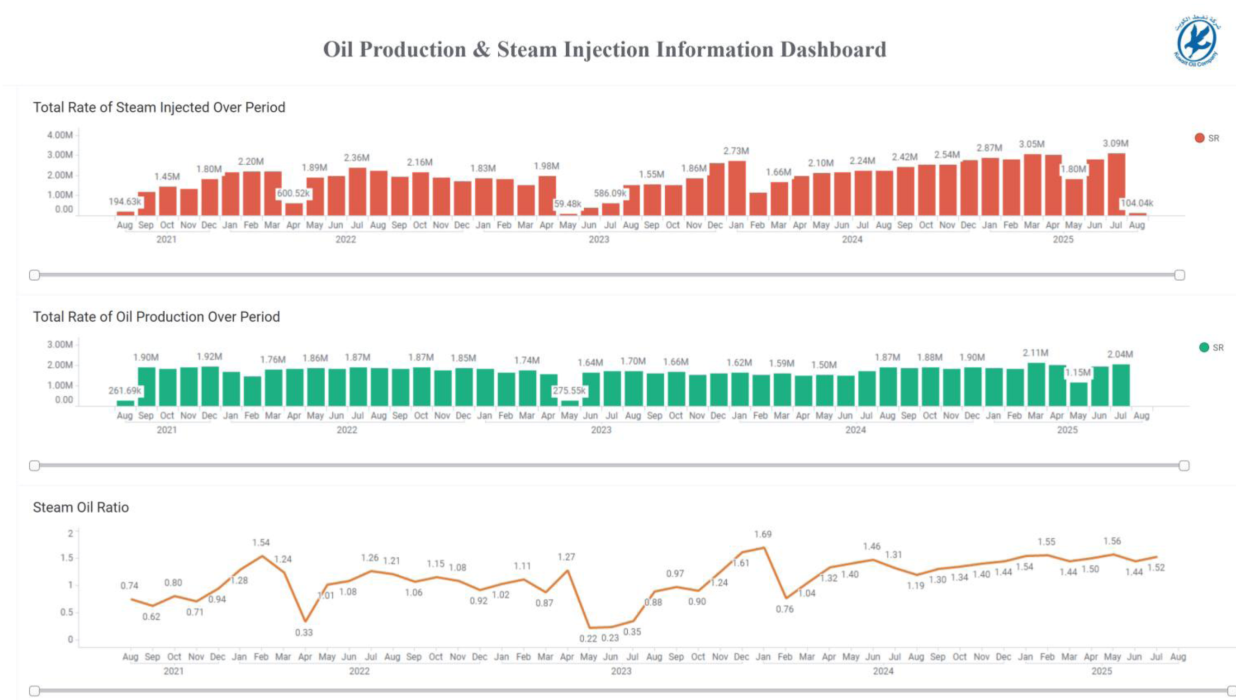

Figure 6—Oil–Steam Overlay & SOR Tracking

Integrated Operational Intelligence. Together, these forecasting, spatial, and thermal monitoring features provide a multidimensional view of field performance. By linking geospatial insights with predictive analytics, the dashboard empowers multidisciplinary teams to:

- Detect anomalies earlier,
- Localize issues geographically,
- Make data-driven decisions that improve production performance and reduce operational risk.

These enhancements not only elevate the platform beyond traditional reporting tools but also align with the digital transformation pillars of real-time situational awareness, proactive resource allocation, and continuous performance improvement.

System Integration and Deployment – AVOCET Integration

The successful delivery of real-time, governed analytics in the NKHO Dashboard hinged on robust integration between the AVOCET system and the Spotfire platform. AVOCET acts as the central processing engine—aggregating, validating, and structuring raw field data into operational KPIs—as also implemented in major digital field programs across the Middle East (SLB, 2023). The integration architecture ensures that stakeholders access consistent, traceable, and up-to-date information for decision-making across disciplines.

End-to-End Data Pipeline Design (SIPOC Framework)

To ensure governed analytics and high-integrity data delivery, the dashboard's architecture was developed around a structured SIPOC (Suppliers–Inputs–Processes–Outputs–Customers) framework. This model outlines the full lifecycle of operational data—from raw input sources to actionable insights consumed via the NKHO DSC Dashboard.

The NKHO DSC Dashboard architecture follows a SIPOC (Suppliers–Inputs–Processes–Outputs–Customers) framework to ensure governed, high-integrity data delivery. This structured approach provides full transparency from raw data acquisition to operational intelligence in Spotfire. Data is sourced from multiple production and activity systems, validated and processed in AVOCET, and transformed into standardized KPIs that drive field-level decision-making. Fig. 7 summarizes this end-to-end pipeline architecture, reinforcing how raw operational inputs are transformed into governed, actionable intelligence for fieldwide decision-making.

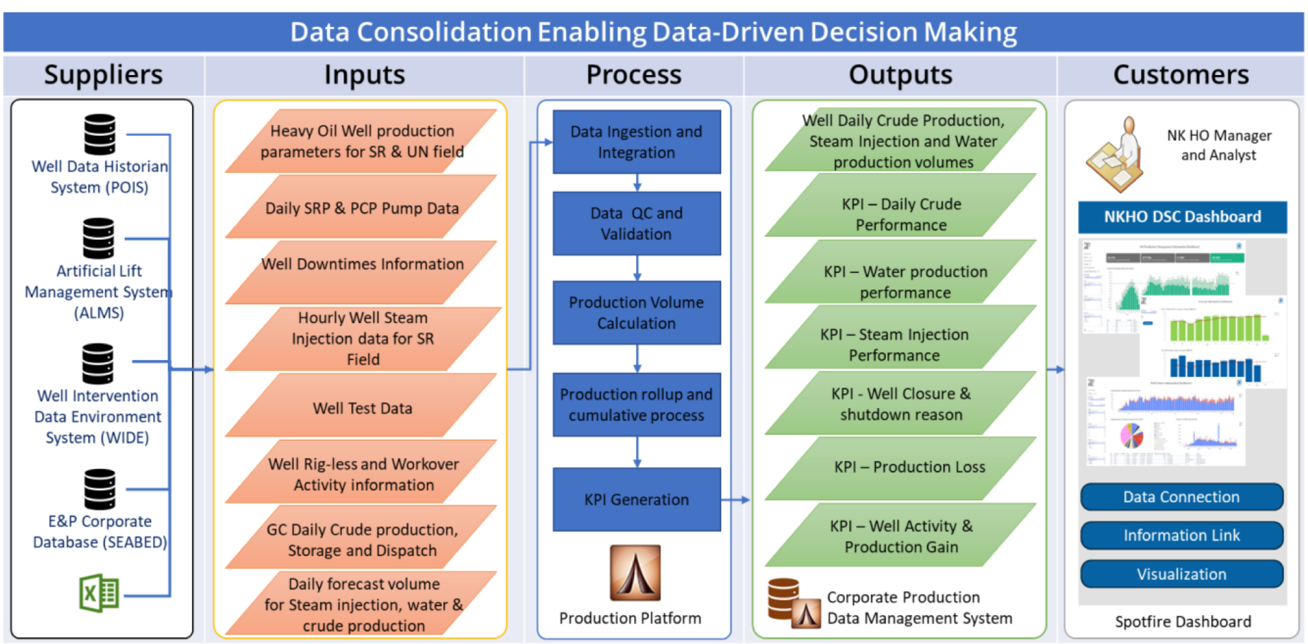

Figure 7—SIPOC Diagram: AVOCET Data Pipeline Enabling Data-Driven Decision Making

Validated Connectivity via Spotfire Information Links

Data is accessed through validated Information Links stored in the centralized KOC Spotfire Library under / E&P-IM Team/Data Connections. These links offer secure, real-time connectivity to AVOCET's BI_USER schema, eliminating the risk of manual data extraction or local inconsistencies. Daily auto-refresh schedules ensure the dashboard reflects the latest field data, with typical report load times under one minute.

Pre-Production QA/QC Protocols

Before deployment to the production environment, dashboards underwent a structured QA/QC process to validate logic, layout, and data accuracy:

- **Draft Review**: Dashboards were reviewed in isolated user folders for functionality, formatting, and completeness.
- **AVOCET Cross-Validation**: Outputs were compared directly against AVOCET backend views to ensure integrity.
- **Stakeholder Sign-Off**: Field teams reviewed KPI outputs for clarity, naming conventions, and unit conversions, resolving any operational discrepancies prior to release.

This rigorous integration and validation approach ensured not only high data fidelity but also widespread user trust, setting the stage for long-term adoption and operational reliance on the NKHO DSC Dashboard.

Results and Conclusions

The deployment of the NKHO Data Standardization and Consistency (DSC) Dashboard has delivered tangible improvements across key operational and digital transformation performance areas. These outcomes were assessed through internal benchmarking methodologies that compared pre- and post-implementation performance baselines.

Quantified Operational Improvements

- **Data Consistency (92%)**: Approximately 92% of critical KPIs are now directly sourced from validated AVOCET data streams. This transition from manual data compilation to automated sourcing has significantly improved traceability, auditability, and stakeholder confidence in reported metrics.
- **Reporting Efficiency (38% Faster)**: Automation of data aggregation and dashboard visualization has reduced average reporting cycle times by 38%. Near real-time access to standardized KPIs now enables faster insight generation and more agile operational response.
- **Intervention Planning Timelines (25% Faster)**: Integrated access to well histories, production trends, and intervention records has shortened planning cycle durations by 25%. This improvement reflects enhanced cross-functional alignment and more efficient field execution readiness.

These improvements underscore the initiative's value in advancing KOC's broader digital transformation goals while enabling more timely, reliable, and data-informed decisions in complex heavy oil operations.

Strategic Use Cases and Productivity Gains

With 72–75 active users across engineering, operations, and IT, the dashboard has evolved into a critical decision-support platform. Automation of repetitive data workflows is estimated to have saved over **5,798 labor hours annually**, translating into both productivity gains and operational cost savings. Previously a minimum of 2-hour daily reporting task is now completed in under 10 minutes via auto-refreshed dashboards —streamlining workload and reducing delays. Key use cases include:

- **Well Closure Analysis**:

  The Well Closure Analysis module provides a comprehensive, multi-dimensional view of closure events across the NKHO. It consolidates three critical perspectives into a single interface: the trend in total closures over time, the categorization of downtime causes (e.g., load limit constraints, mechanical failures, steam-related shutdowns), and the duration of each closure event. This integration enables engineers and operations teams to rapidly pinpoint recurring high-impact issues, quantify their operational impact, and prioritize interventions that deliver the greatest uptime improvement. By linking closure analytics directly to intervention planning, the module supports more targeted maintenance scheduling, reduces average closure duration, and enhances the accuracy of production availability forecasting. Ultimately, this capability transforms closure tracking from a retrospective reporting task into a proactive performance management tool, driving measurable reductions in downtime and optimizing resource allocation (Fig. 8).
- **Production Loss Diagnosis**:

  The Production Loss Diagnosis module rapidly identifies and analyzes the root causes of unplanned production deferments. Interactive trend views break down losses by cause, volume, and timeframe, providing clear visibility into the most significant drivers. This capability enables targeted mitigation and removes delays associated with manual reporting, allowing prompt corrective action to minimize deferred production. Consistent use of these insights helps reduce recurrence, stabilize production, and improve overall asset efficiency.
- **Steam Efficiency Monitoring**:

  The Steam Efficiency Monitoring module continuously tracks the Steam-Oil Ratio (SOR) across fields, patterns, and injection zones. By linking steam injection rates with production volumes, it benchmarks thermal performance and flags high-SOR zones for corrective action. In early 2025, elevated SOR in Pattern B-05 prompted a targeted reallocation, improving recovery efficiency by 12% within one month and enhancing thermal recovery cost-effectiveness.

  These capabilities have not only improved day-to-day field management but have also fostered a cultural shift toward proactive, analytics-driven collaboration across the asset.

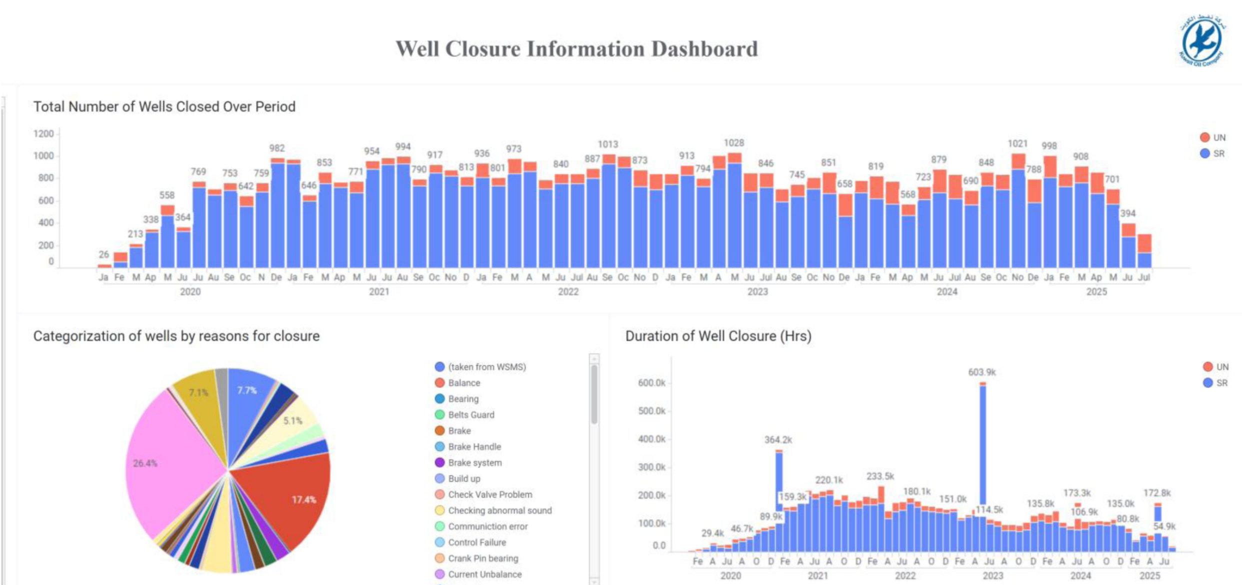

Figure 8—Well Closure Analysis dashboard showing closure trends over time, categorization by root cause, and duration analysis to support targeted intervention planning and downtime reduction

Conclusion

The NKHO DSC Dashboard demonstrates that integrating standardized KPIs into a unified, intelligent analytics platform can deliver measurable improvements in data governance, operational agility, and cross-functional coordination in heavy oil operations. Beyond fulfilling strategic digital transformation objectives, it has established a scalable foundation for future capabilities such as predictive maintenance, advanced production forecasting, and AI/ML-driven optimization. These advancements not only position the platform beyond traditional reporting tools but also align with the core pillars of digital transformation—real-time situational awareness, proactive resource allocation, and continuous performance improvement. The results achieved in North Kuwait offer a replicable blueprint for upstream digital transformation, providing actionable insights for operators seeking to bridge the gap between data complexity and strategic performance in an evolving energy landscape.

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